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Some useful glossary of Ecology

Practical - 1

Introduction

Ecology - The scientific study of the interactions between organisms (biotic factors) and their physical environment (abiotic factors).

Abiotic : Non-living physical and chemical components of an ecosystem, such as sunlight, soil, water and temperature.

Biotic : The living components of an ecosystem, including plants, animals, bacteria and fungi.

Biosphere : The sum of all ecosystems on Earth, the zone of life.

Biome : A large geographical area characterized by specific climate, flora and fauna (eg tundra, tropical rainforest).

Community : All the populations of different species that live and interact in a particular area.

Population : A group of individuals of the same species that live in the same area.

Habitat : The specific natural home or environment of an organism.

Ecological Niche: The functional role and position of a species within an ecosystem, including its habitat, food resources and behavior.

Interactions and Relationships

Symbiosis: A close long-term interaction between two different species.

Commensalism: A type of symbiosis where one species benefits while the other is neither helped nor harmed.

Mutualism: A relationship between two species where both benefit.

Parasitism: A relationship where one organism (the parasite) lives on or in another (the host), causing harm.

Competition: The interaction between species or individuals fighting for the same limited resources (food, water, light).

Predation: An interaction where one organism (predator) hunts and consumes another (prey).

Keystone Species: A species that has a disproportionately large effect on its ecosystem relative to its abundance.

Energy flow and Nutrients

Trophic level: The position an organism occupies in a food chain (e.g. producer, primary consumers).

Autotroph (Producer): An organism that produces its own food, usually through photosynthesis.

Heterotroph (Consumer): An organism that eats other organisms for energy.